

EwE 6.6 release notes

04 December 2017

This document lists important new capabilities of the Ecopath with Ecosim software in the next instalment of EwE, version 6.6. For a full list, please refer to the [EwE change log](#). For a list of known issues with the software, please refer to the [EwE bug tracking system](#).

The most important few features in EwE 6.6 are:

Contaminant tracing

Ecotracer, the contaminant-tracing module of EwE, has been extensively debugged and was partially rewritten. The user interface has not changed significantly. The following paper described and utilized the new Ecotracer:

Walters, W.J., and Christensen, V. (2018). Ecotracer: analyzing concentration of contaminants and radioisotopes in an aquatic spatial-dynamic food web model. *Journal of Environmental Radioactivity* 181, 118–127.

Explicit accounting for discards

Through the DiscardLess project, Ecopath, Ecosim and Ecospace have been given features to better account for discarding. MS in preparation.

- In Ecopath, computed discards are more clearly shown in new fleet-based output forms
- In Ecosim, a few things have happened.
 - Absolute discard proportions and Absolute Discard mortality rates can be driven through new time series types 10 and 11, respectively. Ecosim can also be fitted through time series of Landings and Discards, new types 12 and 13. These new time series apply to a fleet (first target) and a functional group (second target)
 - The Ecosim group and All fits plots show the new time series
 - Ecosim has also gained a new Fleet plots, where Ecosim results are displayed by fleet
- In Ecospace, spatial discards are computed and can be explored.

Advection

The Ecospace advection model has been replaced; it never really worked in EwE versions 6 up to now. You can either enter in monthly winds from which currents will be calculated, or you can directly enter monthly current patterns. Monthly current patterns are stored with the scenario.

The spatial temporal framework (available through projects) can drive current patterns, too.

Spatial temporal forcing

The spatial-temporal data framework can drive a range of new Ecospace layers, such as habitat maps, MPA maps, and currents.

Flow diagrams

The Ecopath flow diagram now includes display of fleets, and the layout of a diagram is stored in the EwE model. This means that you will no longer lose the layout of your diagrams.

Plug-ins

A range of new plug-ins is added to EwE 6.6, such as Merge and split groups (Geomar, Germany), Management Strategy Evaluation (Cefas, UK), Ecosampler (EII, Spain; MS submitted), Ecospace transect extraction (Cefas, UK), and MPA dynamics (UBC, Canada). All plug-ins install their documentation alongside the EwE User Guide.

Other additions

- EwE models can be exported to .eiixml format from the File menu. XML based model data is useful for running EwE on machines where Access is not available. Note that EwE treats XML models as read-only; models will have to be built and fine-tuned using the regular Access-based model files, and should only be exported to XML when finished.
- Pedigree definitions can be imported from other EwE models.
- Ecosim allows forcing of Other mortality rates (M0) through forcing functions. This functionality is experimental, and may be revised in future releases. There is also already significant interest to bring this functionality to Ecospace, somehow.
- Monte Carlo can now save its data in different formats. Rather than only storing results in one massive CSV file, users can opt to store results in one CSV file per result. This new format is easier to read and explore in Excel, and is easier to process through scripts, but fills up your hard drive faster with smaller files.
- The 'Change shape' interface has been overhauled, and new shape types can be added through plug-ins.
- More EwE components have been made compliant with the Autosave system, which facilitates output collection and running parameter uncertainty assessments through Ecosampler.

Known issues

Please refer to the [EwE bug tracking system](#) to see which issues have been reported but that yet have not been addressed.

The EwE user guide is still hopelessly outdated, but we started a project to rewrite the user guide following the Primer with Permanova model. The new manual will be issued as separate chapters, where science and software usage instructions are clearly separated.

Acknowledgements

EwE 6.6 has been developed through user support and co-development contracts with many institutes, thank you all for supporting the EwE approach.

Development of EwE is coordinated through Ecopath International Initiative Research Association (EII) in Barcelona, Spain.

Be involved

EwE is entirely developed through its user community. As we have not a drop of core funding, we need your involvement for EwE to continue and grow. This means a few different things:

- **If you find a bug, please let us know.** Even if we cannot promise that we will fix it right away (we may need funding), it is essential that known issues are known by the community
- **If you have a great idea for new functionality, let us know. If you have means to fund the development of the new features, even better.** Let us build the new functionality with the

EwE core team, and after you have published the new work, changes to EwE are released to the community with a new version of the software.

- **If you have questions about EwE, try our online resources** such as Facebook or the EwE users forum. We encourage you also to provide help to others.
- **If you need expert help applying EwE to a project, consider taking out a support contract.** One-on-one helps you making the best use of EwE, and helps the EwE approach in return.
- **If you need training, let us know.** Several institutes connected to the Ecopath Research and Development Consortium (and beyond) organize courses, and we are always happy to discuss organizing specific courses on demand.

Contact information

Software development and bugs:	Dev team hotline ; EwE bug tracking website ; Co-development website ; Development website .
User support and training:	User support hotline ; User support website .
EwE community:	Ecopath website ; Ecopath user's forum ; Ecopath on Facebook .